

Pressure and Altitude

Student sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows **Pressure** (in hPa), **Calc. altitude** (in m) and **Temperature**. You start with **Start**, move on with the green button at the bottom right, go back with **Back**, and the last green button reads **Finish**.

You wait 1–2 minutes on every landing, until the number stops drifting. The table on screen does not fill itself in — you write it here. The altitude on the card can be badly wrong; the difference between two levels is not. Nobody reads a screen while walking: the box is carried, it is read standing still, on a landing.

2. What I do on my own

Building: _____ Weather today: _____ Box: _____

First I predict. I do not change the prediction after I measure.

1. At the top, the pressure will be ☐ bigger ☐ smaller ☐ the same as at the ground floor, by _____ hPa,

because _____

2. I would know I was wrong if _____

First I measure the stairwell, with the tape measure. I do not guess. One step riser = _____ cm, steps between two floors = _____, one storey = _____ m, storeys climbed = _____, so the true height = _____ m

Now I measure. The time is written every time — without it I cannot correct for the weather drift.

Level	Time	P (hPa)	App tude (m)	T (°C)
Ground floor (mean)				
Floor 1				
Floor 2				
Top floor (mean)				
Ground floor (re- turn)				

The check. Two readings at the ground floor and two at the top, 30 s apart; enter the mean in the table.

Ground floor: _____

spread (max – min) = _____ hPa

Top floor: _____

spread = _____ hPa

Difference between levels = _____ hPa. If the spread exceeds it, the difference is **not detectable**, not non-existent: ☐ detectable ☐ not detectable

The return is not optional. Weather drift = (return) – (first reading) = _____ hPa. Do I trust the result? ☐ YES ☐ NO

The rule: for every 10 m you climb, the pressure falls by 1.2 hPa (that is, 0.12 hPa per metre).

Conversion, already worked out: 0.4 hPa → 3 m 0.8 → 7 m 1.2 → 10 m 1.6 → 13 m
2.0 → 17 m

ΔP = ground floor _____ – top _____ = _____ hPa $\Delta h = \Delta P / 0.12 =$ _____ m

From the tape measure: _____ m deviation = _____ m as a percentage = deviation / tape measure $\times 100 =$ _____ %

The graph. Pressure (hPa) on the vertical axis, height from the tape measure on the horizontal axis. **The pressure axis does not start at zero.**

top: _____

bottom: _____

Height above the ground floor (m), from 0 to _____ — ten divisions

Gradient, from two points *on the line*: _____ hPa/m; theory gives –0.12 hPa/m.

3. The conclusion I write

I predicted that, and the data show that _____

The altitude on screen is wrong, but the difference is good, because _____

What limited me most was _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.
